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CHAPTER 1 • NUMBER • TOPIC 8

Fractions, Decimals & Percentages

Three ways to write the same number.
The exam's favourite: turn a recurring decimal into a fraction with algebra — make the tails match, then subtract.



$$\frac{3}{10} = 0.3 =$$

30%**5**

VARIANTS

12

PAST-PAPER QUESTIONS

34

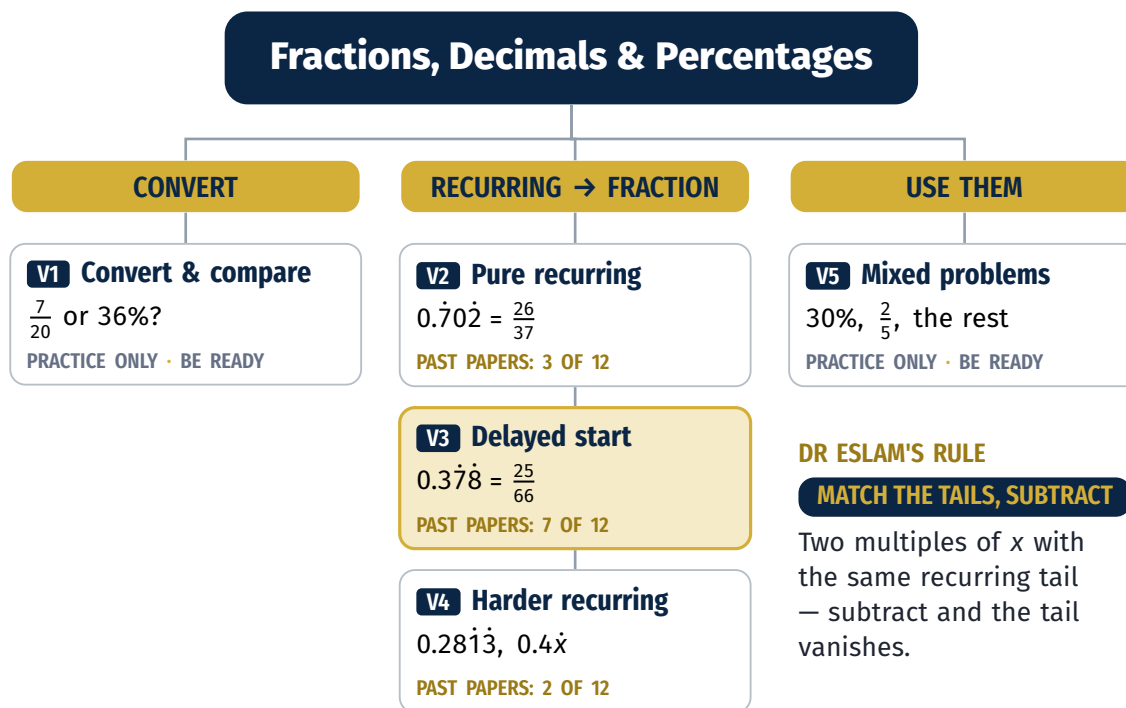
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Convert & compare**V2** Recurring: pure**V3** Recurring: delayed**V4** Recurring: harder**V5** Mixed problems

TOPIC MAP • EVERY VARIANT ON ONE PAGE

FDP at a glance



Your notes


★ How often does each variant come up?

12 past-paper questions (Jan 2020 → Nov 2025); 34 practice parts — grouped by type.

	PAST PAPERS (OUT OF 12)		PRACTICE PARTS
V1 Convert & compare	● ● ● ● ● ● ● ● ● ● ● ●	0	6
V2 Pure recurring	● ● ● ● ● ● ● ● ● ● ● ●	3	4
V3 Delayed start	● ● ● ● ● ● ● ● ● ● ● ●	7	13
V4 Harder recurring	● ● ● ● ● ● ● ● ● ● ● ●	2	1
V5 Mixed problems	● ● ● ● ● ● ● ● ● ● ● ●	0	10

WHAT THIS TELLS YOU Every one of the 12 past-paper questions is “Use algebra to show that $0.\dots = \frac{a}{b}$ ” — 2 marks, almost every series. Learn the subtraction method and it's yours.

BEFORE YOU START

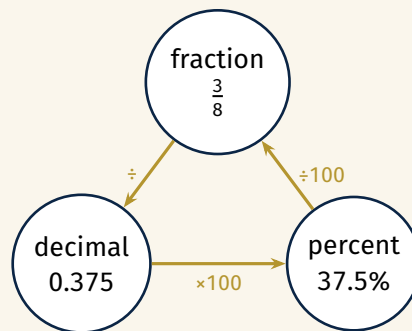
The big idea



One number, three outfits.

$\frac{3}{8} = 0.375 = 37.5\%$. Fraction $\rightarrow$
 decimal: **top $\div$ bottom**. Decimal $\rightarrow$
 %: $\times 100$.

A **recurring** decimal repeats for ever.
 A dot shows the repeating digits:
 $0.3\dot{7}8 = 0.378787 \dots$



Recurring notation

One dot

$0.\dot{5} = 0.5555 \dots$
 that digit repeats

Two dots

$0.\dot{7}0\dot{2} = 0.702702 \dots$
 the block from dot to dot repeats

Delayed

$0.3\dot{7}8 = 0.3787878 \dots$
 the 3 does **not** repeat



Which powers of 10?

Block of 1

$10x - x = 9x$
 $0.\dot{5}$

Block of 2

$100x - x = 99x$
 $0.\dot{4}\dot{5}$; delayed:
 $1000x - 10x = 990x$

Block of 3

$1000x - x = 999x$
 $0.\dot{7}0\dot{2}$



CALCULATOR FIRST, THEN CHECK

Calculator first: type the fraction you must reach ($26 \div 37$) — it shows $0.702702 \dots$, so you know the target is right. Then write the algebra: the calculator alone scores 0 on “use algebra to show”.

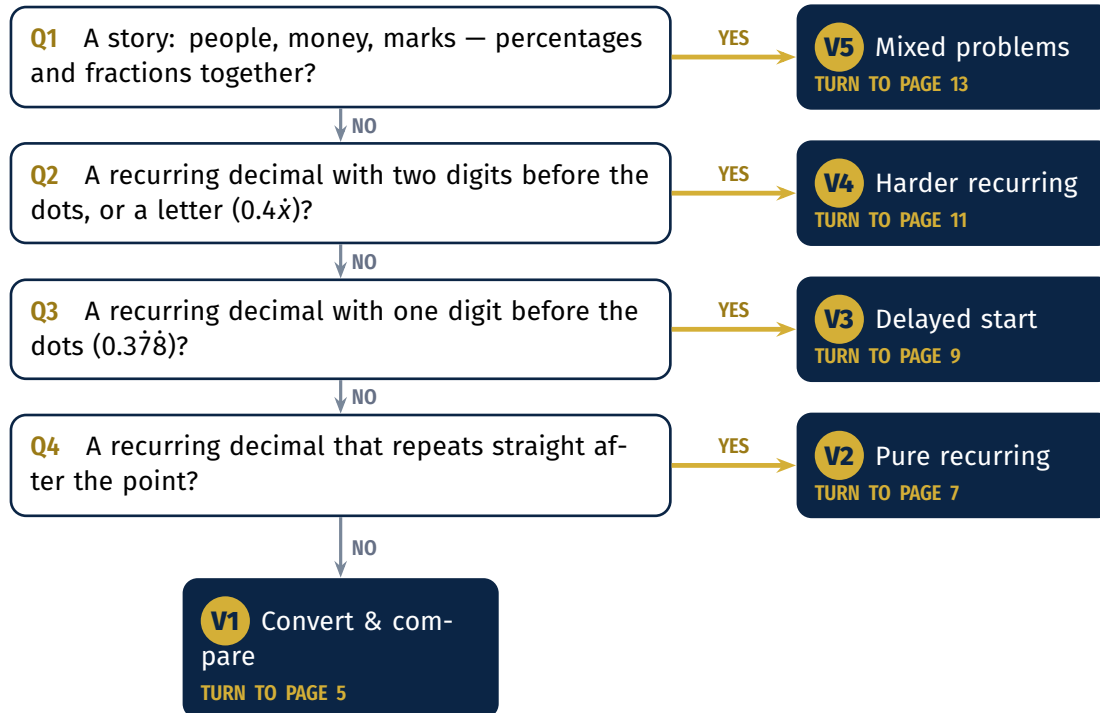
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Convert & Compare

Write everything in the same form (usually decimals or %), then compare or order.

SEEN IN PAST PAPERS

0 / 12

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 6

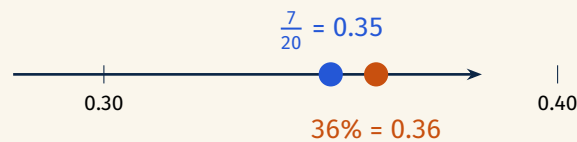
Your notes

? What is it?

In plain words

You can't compare $\frac{7}{20}$ and 36% as they are. Change them to the **same form**: $\frac{7}{20} = 0.35 = 35\%$. Now $35\% < 36\%$.

For ordering, write every number as a decimal with the **same number of places**.



IT LOOKS LIKE

 $\frac{7}{20}$ or 36%

0.6

 $\frac{5}{8}$

63%

How do I do it?

1 Choose a form

Decimals are easiest to compare.

2 Convert

Fraction: top $\div$ bottom. %: $\div 100$.

3 Line up

Same number of decimal places: 0.600, 0.625, 0.630.

4 Answer

In the **original** form the question used.

SAME FORM, THEN COMPARE

Write the original numbers in your answer.

V1 Convert & Compare

VARIANT 1 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

“Which is bigger? Show your working” → **Convert both to decimals or %**

“Write in order of size” → **Decimals with the same places**

“as a recurring decimal” → **Divide; put dots on the repeating block**

WORKED EXAMPLE DR ESAM

Compare • 2 marks

Which is bigger: $\frac{7}{20}$ or 36%? Show your working.

STEP 2 $\frac{7}{20} = 7 \div 20 = 0.35 = 35\%$ ◀ convert

STEP 3 $35\% < 36\%$ ◀ compare

ANSWER 36% is bigger

✓ $0.35 < 0.36$ ✓**WATCH OUT**

× “ $\frac{7}{20}$ ”, because $20 > 7$ ”

✓ **convert first: $35\% < 36\%$**

Compare values, not the digits you can see.

EXAM TIP

Answer with the numbers as the question wrote them: “36% is bigger”.

Check yourself

calculator first • answers on page 15

1.1 Dr Esam

Write $\frac{3}{8}$ as a decimal
and as a percentage.

Answer:

1.2 Practice • Easy

Write $\frac{5}{9}$ as a recurring
decimal.

Answer:

1.3 Dr Esam

Order, smallest first: $\frac{5}{8}$,
63%, 0.6.

Answer:



VARIANT 2

Recurring Decimals: Pure

The block repeats straight after the point:
 $10^n x - x$, where n is the block length.

SEEN IN PAST PAPERS

3 / 12

PRACTICE QUESTIONS: 4

3 / 12

 Your note

Came up in: Jan 2021 P1HR Q15(a) · Jun 2024 P2HR Q18 · Jun 2025 P2H Q18

What is it?

In plain words

Call the decimal x . The block 702 has **3 digits**, so multiply by 1000: the tail lines up with x 's tail.

Subtract: $1000x - x = 999x$ and the endless tail **cancels**. Then divide and simplify.

$$1000x = 702.702702 \dots$$
$$-x = 0.702702 \dots$$

TAILS CANCEL

$$999x = 702$$

IT LOOKS LIKE

 $0.\dot{7}0\dot{2}$ $0.\dot{3}0\dot{6}$

4.57

How do I do it?

1 Name it

$x = 0.702702 \dots$ (write a few repeats).

2 Shift the block

Block of 3 $\rightarrow 1000x = 702.702702 \dots$

3 Subtract

$$1000x - x = 999x = 702.$$

4 Simplify

$$x = \frac{702}{999} = \frac{26}{37} - \text{the given fraction.}$$

BLOCK LENGTH = NUMBER OF ZEROS 2 digits $\rightarrow$ 100, 3 digits $\rightarrow$ 1000.

V2 Recurring Decimals: Pure

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

"Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$ " → $1000x - x = 999x$

A whole number in front: $4.\dot{5}\dot{7}$ → Do $0.\dot{5}\dot{7}$, then add the 4

(2 marks) → The subtraction line is M1, the fraction A1

 WORKED EXAMPLE **PAST PAPER**

June 2025 · Paper 2H · Q18 · 2 marks

Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

STEP 1 $x = 0.702702 \dots$ ◀ name it

STEP 2 $1000x = 702.702702 \dots$ ◀ block of 3: $\times 1000$

STEP 3 $999x = 702$ ◀ subtract: tails cancel

STEP 4 $x = \frac{702}{999} = \frac{26}{37}$ ◀ simplify ($\div 27$)

ANSWER $0.\dot{7}0\dot{2} = \frac{26}{37}$ (shown)

✓ $26 \div 37 = 0.702702 \dots$ ✓

 WATCH OUT

✗ $100x - x$ for a 3-digit block

✓ $1000x - x$

The tails only line up when you shift by the whole block.

 EXAM TIP

Write the decimals one under the other with the repeats lined up — the examiner looks for that.

 Check yourself

calculator first · answers on page 15

2.1 Jun 2024 P2HR

Use algebra to show that $0.\dot{3}0\dot{6} = \frac{34}{111}$.

Answer:

2.2 Jan 2021 P1HR

Use algebra to show that $4.\dot{5}\dot{7} = 4\frac{19}{33}$.

Answer:

2.3 Dr Eslam

Use algebra to show that $0.4\dot{5} = \frac{5}{11}$.

Answer:



VARIANT 3 **MOST COMMON IN THE EXAM**

Recurring Decimals: Delayed Start

A digit before the repeat: make two numbers whose tails start right after the point.

SEEN IN PAST PAPERS

● ● ● ● ● ● ● ● ● ● 7 / 12

PRACTICE QUESTIONS: 13



Your notes

Came up in: Jan 2020 P1HR Q13(a) · Jun 2022 P2HR Q17 · May 2023 P1HR Q16 · Nov 2023 P2H Q15(a) · Nov 2024 P2H Q15 · Jun 2025 P1HR Q16 · Nov 2025 P2H Q16

What is it?

In plain words

In $0.3\dot{7}\dot{8}$ the 3 does **not** repeat. Make **two** multiples of x whose tails both start right after the point:

$$10x = 3.787878 \dots \text{ and}$$
$$1000x = 378.787878... \text{ — subtract:}$$
$$990x = 375.$$

IT LOOKS LIKE

0.378

 $0.\dot{3}\dot{4}\dot{5}$
$$0.1\dot{6}$$

$$1000x = 378.787878 \dots$$

$$-10x = 3.787878\dots$$

SAME TAIL

$$990x = 375$$

How do I do it?

1 Name it

$$x = 0.3787878 \dots$$

2 Move past the delay

$10x = 3.787878... - \text{the repeat now starts at the point.}$

3 Shift one block more

Block of 2: $1000x = 378.787878 \dots$

4 Subtract, simplify

$$990x = 375, \quad x = \frac{375}{990} = \frac{25}{66}.$$

TWO NUMBERS, SAME TAIL Then subtract.

V3 Recurring Decimals: Delayed Start

VARIANT 3 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

"Use algebra to show that $0.3\dot{7}\dot{8} = \frac{25}{66}$ " → $1000x - 10x = 990x$

One recurring digit after a delay: $0.1\dot{6}$ → $100x - 10x = 90x$

The answer must be "in its simplest form" → **Divide top and bottom by the HCF**

 WORKED EXAMPLE **PAST PAPER**

November 2025 · Paper 2H · Q16 · 2 marks

Use algebra to show that $0.3\dot{7}\dot{8} = \frac{25}{66}$

STEP 1 $x = 0.3787878 \dots$

◀ name it

STEP 2 $10x = 3.787878 \dots$

◀ past the delay

STEP 3 $1000x = 378.787878 \dots$

◀ one more block

STEP 4 $990x = 375 \Rightarrow x = \frac{375}{990} = \frac{25}{66}$ ◀ subtract, simplify ($\div 15$)**ANSWER** $0.3\dot{7}\dot{8} = \frac{25}{66}$ (shown)✓ $25 \div 66 = 0.37878 \dots$ ✓ **WATCH OUT**✗ $1000x - x = 999x$ ✓ $1000x - 10x = 990x$

$1000x$ and x have different tails here — the 3 gets in the way.

 EXAM TIP

7 of the 12 past-paper questions. Write both lines one above the other so the tails visibly match.

 Check yourself

calculator first · answers on page 15

3.1

Jun 2022 P2HR

Use algebra to show that $0.3\dot{4}\dot{5} = \frac{19}{55}$.

Answer:

3.2

Jun 2025 P1HR

Use algebra to show that $0.6\dot{1}\dot{2} = \frac{101}{165}$.

Answer:

3.3

Dr Eslam

Use algebra to show that $0.1\dot{6} = \frac{1}{6}$.

Answer:



VARIANT 4

VARIANT 4

Recurring Decimals: Harder

Two digits before the repeat, or a letter:
same idea, bigger shifts.

SEEN IN PAST PAPERS

● ● ● ● ● ● ● ● ● ● 2 / 12
PRACTICE QUESTIONS: 1



Your notes

Came up in: May 2021 P1H Q16 • Nov 2021 P1H Q18

What is it?

In plain words

0.2813: two digits (28) before the repeat. 100x moves past them; 10 000x moves one more block of 2.

0.4x: treat x as a digit —
 $10n = 4.xxx\dots$, $100n = 4x.xxx\dots$ —
the same method.

IT LOOKS LIKE

$$0.28\dot{1}\dot{3}$$
 $0.4\dot{x}$

$$10\,000x = 2813.1313\dots$$

$$-100x = 28.1313\dots$$

SAME TAIL

$$9900x = 2785$$

How do I do it?

1 Past the delay

Delay of 2 digits $\rightarrow 100\times$.

2 One more block

Block of 2 $\rightarrow 10\,000x$.

3 Subtract

$$9900x = 2785.$$

4 Simplify

$$\frac{2785}{9900} = \frac{557}{1980} (\div 5).$$

COUNT THE DELAY, COUNT THE BLOCK

They decide the two multipliers.

V4 Recurring Decimals: Harder

VARIANT 4 · SEE IT DONE, THEN TRY


 Your notes
 WHEN YOU SEE → YOU DOTwo digits before the repeat: $0.28\dot{1}\dot{3}$ → $10\,000x - 100x = 9900x$ “ $0.4\dot{x}$... find, in terms of x , ... as a fraction” → Use a new letter n for the decimal

“in its simplest form” → Cancel common factors

 WORKED EXAMPLE **PAST PAPER**

May 2021 · Paper 1H · Q16 · 2 marks

Use algebra to show that the recurring decimal $0.28\dot{1}\dot{3} = \frac{557}{1980}$ **STEP 1** $x = 0.2813131 \dots$ ◀ name it**STEP 2** $100x = 28.131313 \dots$, $10\,000x = 2813.131313 \dots$ ◀ past the delay; one more block**STEP 3** $9900x = 2785$ ◀ subtract**STEP 4** $x = \frac{2785}{9900} = \frac{557}{1980}$ ◀ simplify ($\div 5$)**ANSWER** $0.28\dot{1}\dot{3} = \frac{557}{1980}$ (shown)✓ $557 \div 1980 = 0.28131 \dots$ ✓ **WATCH OUT**✗ using x for the decimal when the question uses x as a digit✓ call the decimal n

One letter for one thing — otherwise the algebra breaks.

 EXAM TIP

Same three lines every time: name it, two shifted copies, subtract.

 Check yourself

calculator first · answers on page 15

4.1 Nov 2021 P1H x is a whole number,
 $1 \leq x \leq 9$. Write $0.4\dot{x}$ as
a fraction in terms of x .

Answer:

4.2 Practice · HardWrite $0.28\dot{1}$ as a
fraction in its simplest
form.

Answer:

4.3 Dr EslamWrite $0.12\dot{3}$ as a
fraction in its simplest
form.

Answer:



VARIANT 5

Mixed Problems

Fractions and percentages of the same whole: change them all to amounts first.

SEEN IN PAST PAPERS

0 / 12

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 10



Your notes

❓ What is it?

In plain words

“30% walk, $\frac{2}{5}$ take the bus, the rest drive”: work out

each part as an amount of the whole, then the rest is whole minus the parts.

“Of the rest”: the second fraction acts on what is **left**.

IT LOOKS LIKE

30%

 $\frac{2}{5}$

the rest

“of the remaining”

200 people

30%: 60	$\frac{2}{5}$: 80	rest: 60
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How do I do it?

1 The whole

What is 100%? (the total people, money ...).

2 Each part

Amount = fraction or % $\times$ the whole (or of the rest).

3 The rest

Whole – the parts.

4 Answer

As a number, fraction or % — as asked.

AMOUNTS, NOT PERCENTAGES

AMOUNTS, NOT PERCENTAGES You can't add 30% and $\frac{2}{5}$ of different wholes.

V5 Mixed Problems

VARIANT 5 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"... the rest drive. How many drive?" → **Whole – every part**

" $\frac{1}{4}$... then 25% of the remaining" → **Second part of the rest**

"Who did better?" (a mark out of 40 vs a %) → **Change both to %**

WORKED EXAMPLE DR ESAM

Mixed problem • 3 marks

200 people: 30% walk, $\frac{2}{5}$ take the bus and the rest drive. How many drive?

STEP 2 $0.3 \times 200 = 60$, $\frac{2}{5} \times 200 = 80$ ◀ each part

STEP 3 $200 - 60 - 80 = 60$ ◀ the rest

ANSWER 60 people drive✓ $60 + 80 + 60 = 200$ ✓**WATCH OUT**

✗ $30\% + \frac{2}{5} = 70\%$... then confused

✓ **work in amounts: 60 and 80**

Amounts add simply; mixed forms do not.

EXAM TIP

Check: all the parts must add back to the whole.

Check yourself

calculator first • answers on page 15

5.1

Dr Eslam

200 people: 30% walk, $\frac{2}{5}$ take the bus, the rest drive. How many drive?

Answer:

5.2

Dr Eslam

Ali scores 31 out of 40.
Ben scores 76%. Who did better?

Answer:

5.3

Dr Eslam

From £64, spend $\frac{1}{4}$,
then 25% of what is left.
How much is left?

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py).

1.1 0.375, 37.5%

1.2 $0.\dot{5}$

1.3 0.6, $\frac{5}{8}$, 63%

2.1 $999x = 306$,
 $\frac{306}{999} = \frac{34}{111}$

2.2 $99x = 57$,
 $4 + \frac{57}{99} = 4\frac{19}{33}$

2.3 $99x = 45$, $\frac{45}{99} = \frac{5}{11}$

3.1 $990x = 342$,
 $\frac{342}{990} = \frac{19}{55}$

3.2 $990x = 606$,
 $\frac{606}{990} = \frac{101}{165}$

3.3 $90x = 15$, $\frac{15}{90} = \frac{1}{6}$

4.1 $\frac{36 + x}{90}$

4.2 $\frac{253}{900}$

4.3 $\frac{37}{300}$

5.1 60

5.2 Ali (77.5%)

5.3 £36

THE WHOLE TOPIC ON ONE PAGE

FDP cheat sheet


Your notes
V1 Convert & compare

$$\frac{7}{20} = 35\% < 36\%$$

Same form, then compare.

V2 Pure 3/12

$$1000x - x = 999x$$

Block length = zeros.

V3 Delayed 7/12

$$1000x - 10x = 990x$$

Two numbers with the same tail.

V4 Harder 2/12

$$10\,000x - 100x = 9900x$$

Count the delay and the block.

V5 Mixed problems

$$200 - 60 - 80 = 60$$

Work in amounts.

**TOP 3 LOST MARKS**

1. 999x when the start is delayed ×
2. tails not lined up ×
3. fraction not simplified ×

BOTTOM LINE Fraction ÷ → decimal ×100 → % · recurring: name it, two copies with the same tail, subtract, simplify.